

thousand other things. The Indians of New Spain value it greatly, and they commonly have one or several of these trees near their homes to supply their needs. It grows in the fields, and there they cultivate it. Its leaves are wide and thick, with strong, sharp points which they use as fastening pins or sewing needles; they also draw a certain fiber or thread from the leaves.

They cut through the thick trunk when it is tender; there is a large cavity inside, where the sap rises from the roots; it is a liquor which they drink like water, since it is fresh and sweet. When this liquor is boiled it turns into a kind of wine, and if it is left to sour it becomes vinegar. But when boiled for a longer time it becomes like honey, and cooked half as long it turns into a healthful syrup of good flavor, superior in my judgment to syrup made of grapes. Thus they boil different substances from this sap, which they obtain in great quantity, for at a certain season they extract several *azumbres* a day.

2. SPAIN'S CONTRIBUTIONS TO NEW WORLD AGRICULTURE

The colonial era saw a notable exchange of agricultural gifts between the Old and the New Worlds. The Spanish crown displayed much solicitude for the agricultural development of the Indies and paid particular attention to the shipping of trees, plants, seeds, and agricultural implements of all kinds. Father Acosta gives an account of the transit of Spanish plants to America and of the rapid rise there of a commercial agriculture producing wine, wheat, sugar, and other products.

The Indies have been better repaid in the matter of plants than in any other kind of merchandise; for those few that have been carried from the Indies into Spain do badly there, whereas the many that have come over from Spain prosper in their new homes. I do not know whether to attribute this to the excellence of the plants that go from here or to the bounty of the soil over there. Nearly every good thing grown in Spain is found there; in some regions they do better than in others. They include wheat, barley, garden produce and greens and vegetables of all kinds, such as lettuce, cabbage, radishes, onions, garlic, parsley, turnips, carrots, eggplants, endive, salt-wort, spinach, chickpeas, beans, and lentils—in short, whatever grows well here, for those who have gone to the Indies have been careful to take with them seeds of every description. . . .

The trees that have fared best there are the orange, lemon, citron, and others of that sort. In some parts there are already whole forests and groves

of orange trees. Marveling at this, I asked on a certain island who had planted so many orange trees in the fields. To which they replied that it might have happened that some oranges fell to the ground and rotted, whereupon the seeds germinated, and, some being borne by the waters to different parts, gave rise to these dense groves. This seemed a likely reason. I said before that orange trees have generally done well in the Indies, for nowhere have I found a place where oranges were not to be found; this is because everywhere in the Indies the soil is hot and humid, which is what this tree most needs. It does not grow in the highlands; oranges are transported there from the valleys or the coast. The orange preserve which is made in the islands is the best I have ever seen, here or there.

Peaches and apricots also have done well, although the latter have fared better in New Spain. . . . Apples and pears are grown, but in moderate yields; plums give sparingly; figs are abundant, chiefly in Peru. Quinces are found everywhere, and in New Spain they are so plentiful that we received fifty choice ones for half a *real*. Pomegranates are found in abundance, but they are all sweet, for the people do not like the sharp variety. The melons are very good in some regions, as in Tierra Firme and Peru. Cherries, both wild and cultivated, have not so far prospered in the Indies. . . . In conclusion, I find that hardly any of the finer fruits is lacking in those parts. As for nuts, they have no acorns or chestnuts, nor, as far as I know, have any been grown over there until now. Almonds grow there, but sparingly. Almonds, walnuts, and filberts are shipped there from Spain for the tables of epicures. I have not seen any medlars or services, but those do not matter. . . .

By profitable plants I mean those plants which not only yield fruit but bring money to their owners. The most important of these is the vine, which gives wine, vinegar, grapes, raisins, verjuice, and syrup—but the wine is the chief concern. Wine and grapes are not products of the islands or of Tierra Firme; in New Spain there are vines which bear grapes but do not yield wine. The reason must be that the grapes do not ripen completely because of the rains which come in July and August and hinder their ripening; they are good only for eating. Wine is shipped from Spain and the Canary Islands to all parts of the Indies, except Peru and Chile, where they have vineyards and make very good wine. This industry is expanding continually, not only because of the goodness of the soil, but because they have a better knowledge of winemaking.

The vineyards of Peru are commonly found in warm valleys where they have water channels; they are watered by hand, because rain never falls in the coastal plains, and the rains in the mountains do not come at the proper time. The vineyards have increased so far that because of them the tithes of the churches are now five and six times what they were twenty years ago. The valleys most fertile in vines are Victor, near Arequipa; Yca, hard by Lima; and Caracaro, close to Chuquibabo. The wine that is made there is shipped to Potosí

and Cuzco and various other parts, and it is sold in great quantities, because since it is produced so abundantly it sells at five or six ducats the jug, or *arroba*, whereas Spanish wine (which always arrives with the fleets) sells for ten and twelve. . . . The wine trade is no small affair, but does not exceed the limits of the province.

The silk which is made in New Spain goes to other provinces—to Peru, for example. There was no silk industry before the Spaniards came; the mulberry trees were brought from Spain, and they grow well, especially in the province called Misteca, where they raise silkworms and make good taffetas; they do not yet make damasks, satins, or velvets, however.

The sugar industry is even wider in scope, for the sugar not only is consumed in the Indies but is shipped in quantity to Spain. Sugar cane grows remarkably well in various parts of the Indies. In the islands, in Mexico, in Peru, and elsewhere they have built sugar mills that do a large business. I was told that the Nasca [Peru] sugar mill earned more than thirty thousand pesos a year. The mill at Chicama, near Trujillo [Peru], was also a big enterprise, and those of New Spain are no smaller, for the consumption of sugar and preserves in the Indies is simply fantastic. From the island of Santo Domingo, in the fleet in which I came, they brought eight hundred and ninety-eight chests and boxes of sugar. I happened to see the sugar loaded at the port of Puerto Rico, and it seemed to me that each box must contain eight *arrobas*. The sugar industry is the principal business of those islands—such a taste have men developed for sweets!

Olives and olive trees are also found in the Indies, in Mexico, and in Peru, but up to now they have not set up any mills to make olive oil. Actually, it is not made at all, for they prefer to eat the olives, seasoning them well. They find it unprofitable to make olive oil, and so all their oil comes from Spain.

3. THE POTOSÍ MINE

Spain's proudest possession in the New World was the great silver mine of Potosí in Upper Peru (present-day Bolivia), whose flow of treasure attained gigantic proportions between 1579 and 1635. More than any other colonial resource, the fantastic wealth of Potosí captivated the Spanish imagination. The following selection gives some account of this wealth and of mining practices in the late sixteenth century.

It appears from the royal accounts of the House of Trade of Potosí, and it is affirmed by venerable and trustworthy men, that during the time of the

government of the licentiate Polo, which was many years after the discovery of the hill, silver was registered every Saturday to the value of 150 to 200,000 pesos, of which the King's fifth (*quinto*) came to 30 to 40,000 pesos, making a yearly total of about 1,500,000 pesos. According to this calculation, the value of the daily output of the mine was 30,000 pesos, of which the King's share amounted to 6,000 pesos. One more thing should be noted in estimating the wealth of Potosí; namely, that accounts have been kept of only the silver that was marked and taxed. But it is well known in Peru that for a long time the people of that country used the silver called "current," which was neither marked nor taxed. And those who know the mines well conclude that at that time the bulk of the silver mined at Potosí paid no tax, and that this included all the silver in circulation among the Indians, and much of that in use among the Spaniards, as I could observe during my stay in that country. This leads me to believe that a third—if not one half—of the silver production of Potosí was neither registered nor taxed. . . . [It should also be noted that] although the mines of Potosí have been dug to a depth of two hundred *estados*, the miners have never encountered water, which is the greatest possible obstacle to profitable operations, whereas the mines of Porco, so rich in silver ore, have been abandoned because of the great quantity of water. For there are two intolerable burdens connected with the search for silver: the labor of digging and breaking the rock, and that of getting out the water—and the first of these is more than enough. In fine, at the present time His Catholic Majesty receives on the average a million pesos a year from his fifth of the silver of Potosí, not counting the considerable revenue he derives from quicksilver and other royal perquisites. . . .

The hill of Potosí contains four principal veins: the Rich vein, that of Centeno, the vein called "of Tin," and that of Mendieta. All these veins are in the eastern part of the hill, as if facing the sunrise; there is no vein to the west. These veins run from north to south, or from pole to pole. They measure six feet at their greatest width, and a *palm* at the narrowest point. From these veins issue others, as smaller branches grow out of the arms of trees. Each vein has different mines that have been claimed and divided among different owners, whose names they usually bear. The largest mine is eighty yards in size, the legal maximum; the smallest is four yards. By now all these mines are very deep. In the Rich vein there are seventy-eight mines; they are as deep as one hundred and eighty and even two hundred *estados* in some places. In the Centeno vein there are twenty-four mines. Such are as much as sixty and even eighty *estados* deep, and the same is true of the other veins and mines of that hill. In order to work the mines at such great depths, tunnels (*socavones*) were devised; these are caves, made at the foot of the mountain, that cross it until they meet the veins. Although the veins run north to south, they descend from the top to the foot of the mountain—a distance calculated at more than 1200 *estados*. And by this calculation, although the mines run so deep it is six times

as far again to their root and bottom, which some believe must be extremely rich, being the trunk and source of all the veins. But so far experience has proven the contrary, for the higher the vein the richer it is, and the deeper it runs the poorer the yield. Be that as it may, in order to work the mines with less cost, labor, and risk, they invented the tunnels, by means of which they can easily enter and leave the mines. They are eight feet wide and one *estado* high, and are closed off with doors. With the aid of these tunnels they get out the silver ore without difficulty, paying the owner of the tunnel a fifth of all the metal that is obtained. Nine tunnels have already been made, and others are being dug. A tunnel called "of the Poison" (*del Veneno*), which enters the Rich vein, was twenty-nine years in the making, for it was begun in 1556 (eleven years after the discovery of those mines) and was completed on April 11, 1585. This tunnel crossed the vein at a point thirty-five *estados* from its root or source, and from there to the mouth of the mine was 135 *estados*; such was the depth of that they had to descend to work those mines. This tunnel (called the *Cruero*) is 250 yards in length, and its construction took twenty-nine years; this shows how much effort men will make to get silver from the bowels of the earth. They labor there in perpetual darkness, not knowing day from night; and since the sun never penetrates these places, they are not only always dark but very cold, and the air is very thick and alien to the nature of men. And that is why those who enter there for the first time get seasick, as it were, being seized with nausea and stomach cramps, as I was. The miners always carry candles, and they divide their labor so that some work by day and rest by night and others work at night and rest during the day. The silver ore is generally of a flinty hardness, and they break it up with bars. Then they carry the ore on their backs up ladders made of three cords of twisted cowhide, joined by pieces of wood that serve as rungs, so that one man can climb up and another come down at the same time. These ladders are ten *estados* long, and at the top and bottom of each there is a wooden platform where the men may rest, because there are so many ladders to climb. Each man usually carries on his back a load of two *arrobos* of silver ore tied in a cloth, knapsack fashion; thus they ascend, three at a time. The one who goes first carries a candle tied to his thumb, because, as I mentioned, they receive no light from above; thus, holding with both hands, they climb that great distance, often more than 150 *estados*—a fearful thing, the mere thought of which inspires dread. So great is the love of silver, which men suffer such great pains to obtain.

4. THE COLONIAL FACTORY

The Mexican city of Puebla was a leading industrial center in the colonial period, with numerous workshops (obrajes) producing cotton, woolen, and silk cloth, hats, chi-

naware, and glass. The Englishman Thomas Gage, who visited Puebla not many years after the visit described below, observed that "the cloth which is made in it . . . is sent far and near, and [is] judged now to be as good as the cloth of Segovia, which is the best that is made in Spain."

This city [Puebla] has great workshops in which they weave a quantity of fine cotton cloth, wool frieze, and ribbed silk from which they make a handsome profit because this business is common in these parts. And although the workshop owners are compassionate Christians, in order to supply their workshops with people to make the cloth and ribbed silk, they have persons hired and paid to deceive poor innocents. When they see some Indian from out of town, they lure him to the workshop with tricks or on some pretext like hiring him as a porter and paying him to carry something. When he enters inside the workshop they spring the trap, and the poor soul never again leaves that jail until he dies and is taken away to be buried. In this way they have seized and deceived many an Indian, married with children, forgotten for twenty years or more and for all his life, without his wife or his children knowing where he is, because even though he might want to leave, he cannot, since the city gates are well guarded by watchmen. These Indians are kept busy with carding, spinning, weaving and the other tasks associated with the making of cloth and ribbed silk. And the owners make their profits by these unjust and illicit means.

The Royal Council of the Indies, with holy zeal in the service of Our Lord God, His Majesty, and for the good of the Indians, has tried to remedy the situation through the warrants and ordinances that were sent and continue to be sent for the proper administration of the Indians and for their relief from this great burden and enslavement. And the Viceroy of New Spain [Mexico] has appointed inspectors to visit the workshops and remedy such things. But since most of those charged with these duties are concerned more with their own betterment than with helping the Indians—although it might weigh on their consciences—and since the workshop owners pay them well, the miserable Indians are left in the same slavery. And if some inspectors are carried away by a holy zeal to remedy these abuses when they go to inspect the workshops, the owners hold the workers in some previously designated area, where they hide the Indians against their will so that they are not seen, or found, or able to voice their complaints. This is so common in all the workshops of this city and jurisdiction, and that of Mexico City, and those that build and run the workshops do so without scruples—as if it were not a grave mortal sin.

Antonio Vázquez de Espinosa, *Compendio y descripción de las Indias Occidentales*, 2 vols. (1636), vol. 1, pp. 218-220. Excerpt translated by the editors.